

Literature-Based Cross-Architecture Comparison (Paragraph Format)

1 Introduction

This part is similar to the previous part on Cross Architecture Phenomenon Mapping. The key difference is that the previous section explores more about phenomena (Does XXX happen?), while this section is more about metrics and methodology (How exactly did XXX happen, what were the numbers, and were they a fair comparison?).

Part 1: Cross-Architecture Metric Compilation

1.1 Key Terms

- **Entropy:** The extent of randomness or disorderly in a system. The higher the entropy, the more random the system becomes. Entropy can be modeled using the formula $p(x) \cdot \log_2(p(x))$ where $p(x)$ is the probability distribution of x .
- **Intrinsic Dimension:** The minimum number of variables that needs to be kept to represent a dataset or loss space without losing significant information.
- **Effective Rank:** (Similar to Intrinsic Dimension) A quantitative, often entropy-based measure that characterizes the true intrinsic dimensionality or “complexity” of matrices, operators, distributions, or feature sets encountered. Intrinsic dimension measures the geometric complexity of data (the degrees of freedom required to describe a dataset), while effective rank measures the linear complexity of a system (the number of principal components needed to capture a feature matrix’s variance).
- **Cosine Similarity:** Measures how similar two datapoints are to each other based on their direction.

Entropy (Information Content)

- **Transformer:** Transformers do not monotonically decrease in uncertainty. Transformers balance depth-invariant expansion (where entropy temporarily rises) and pruning (where entropy drops as the model actively narrows down its decision). The shape of the entropy profile is dependent on the model (Ali et al., 2026).
- **Mamba / SSM:** No known research has directly measured change in entropy layer by layer. However, we know that semantic information flows and accumulates naturally over the token sequence [*Forget BIT, It is All about TOKEN*].

- **Diffusion LLM:** No known research has directly measured change in entropy layer by layer. However, we know continuous semantic certainty increases through reverse prediction [*Simple and Effective Masked Diffusion Language Models*].
- **Mixture-of-Experts (MoE):** No unified theory.
- **Comparison Status:** Single-architecture mostly. *Forget BIT* offers a unified semantic framework, but layer-wise entropy profiles are only mapped for Transformers.

Intrinsic Dimension (ID)

- **Transformer:** Compresses from high-dimensional embedding spaces (10^3) down to ≈ 10 -dimensional submanifolds in intermediate layers [*Bridging the Dimensional Chasm*].
- **Mamba / SSM:** Unknown. Current ID tools rely on explicit layer-by-layer structures, making it difficult to measure in state spaces [*Bridging the Dimensional Chasm*].
- **Diffusion LLM:** Unknown.
- **Mixture-of-Experts (MoE):** Unknown. Though routing relies on curved mathematical spaces (Grassmannian manifolds), the intrinsic dimension of the routed tokens is unmeasured [*Grassmannian Mixture-of-Experts: Concentration-Controlled Routing on Subspace Manifolds*].
- **Comparison Status:** Single-architecture. Dimensional chasm theory is currently isolated to Transformers.

Effective Rank

- **Transformer:** Rank mathematically decreases with depth as massive token activations in the residual stream push tokens into compression valleys [*Attention Sinks and Compression Valleys in LLMs are Two Sides of the Same Coin*].
- **Mamba / SSM:** Unknown. It is an open question how representational rank evolves with depth in sequential models [*Attention Sinks and Compression Valleys in LLMs are Two Sides of the Same Coin*].
- **Diffusion LLM:** Unknown.
- **Mixture-of-Experts (MoE):** Unknown.
- **Comparison Status:** Single-architecture. Rank degradation is tightly coupled to pre-norm Transformer mechanics.

Token Interaction

- **Transformer:** Traditional transformers use unbounded dot-product attention ($QK^\top/\sqrt{d}$) which can cause softmax saturation and unstable training. Cosine similarity attention addresses this by applying ℓ_2 normalization to queries (Q) and keys (K), projecting them onto a unit hypersphere:

$$Q'_i = \frac{Q_i}{\|Q_i\|_2} \quad K'_j = \frac{K_j}{\|K_j\|_2}$$

The token interaction matrix is then scaled using a learnable temperature parameter α :

$$\text{Attention}(Q, K, V) = \text{softmax}(\alpha \cdot Q'K'^\top)V$$

This bounds interactions to $[-1, 1]$, stabilizing gradients in massive architectures (Liu et al., 2022) and enabling linear-time computation variants (Mongaras et al., 2024).

- **Mamba / SSM:** Token information is mixed through an input dependent recurrence:

$$h_t = \overline{A}_t h_{t-1} + \overline{B}_t x_t$$

- **Diffusion LLM:** Bidirectional scaled dot-product attention to model token interactions simultaneously across the entire sequence:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V$$

Unlike causal autoregressive models, this non-causal interaction allows every token to look at both left and right contexts simultaneously during each iterative denoising step.

- **Mixture-of-Experts (MoE):** Scaled dot-product attention for token-to-token interactions within the backbone:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V$$

For token-to-expert interaction (routing), they rely on a linear dot-product gating mechanism:

$$G(x) = \text{softmax}(\text{TopK}(x \cdot W_g, k))$$

Tokens interact with gating weights (W_g) via an unconstrained inner product to determine which k experts receive the token.

- **Comparison Status:** Different architectures have different models of token interaction.

Part 2: Comparison Narratives & Methodological Differences

Narrative 1: The Measurement of Entropy is Fractured

The literature agrees that tokens compress information, but the methodology for measuring this entropy is fractured by architecture. In Transformers, tools like “Entropy-Lens” are observational—they probe the residual stream to plot a 1D entropy profile layer-by-layer [*Entropy-Lens: Uncovering Decision Strategies in LLMs*]. In contrast, MoE literature uses entropy functionally rather than observationally; DynaMoE actively computes a token’s confidence threshold (entropy) to dynamically allocate expert compute on the fly [*DynaMoE: Dynamic Token-Level Expert Activation with Layer-Wise Adaptive Capacity for Mixture-of-Experts Neural Networks*]. There is no direct comparative study measuring the Shannon entropy of a Mamba hidden state against a Transformer residual stream at equivalent depths.

Narrative 2: Dimensionality relies on Discrete Layers

Transformers exhibit a “Dimensional Chasm,” collapsing 10^3 parameters into ≈ 10 -dimensional semantic submanifolds [*Bridging the Dimensional Chasm*]. However, the methodological limitation here is severe: this geometric analysis heavily relies on the explicit residual stream and the discrete layer-by-layer structure of Transformers [*Bridging the Dimensional Chasm*]. Because Diffusion models iterate temporally rather than spatially, and Mamba models mediate through a continuous state space, the existing tools to measure Intrinsic Dimension simply do not port over, leaving a massive blank spot in cross-architecture geometric comparison.

Narrative 3: Unrolling the Black Box of Token Interaction

Historically, comparing the pairwise similarity of tokens in a Transformer to those in Mamba was impossible because Mamba squashes everything into a compressed 1D hidden state. The methodological breakthrough comes from LaTIM, which uses dynamical systems math to “unroll” Mamba’s sequential states into extractable, pairwise interaction scores [*LaTIM: Measuring Latent Token-to-Token Interactions in Mamba Models*]. This is one of the few areas where direct, apples-to-apples comparison is now methodologically possible, proving that Mamba structurally mimics Transformer attention implicitly.